Curriculum Vitae

MARILYN JUDITH REGO

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EDUCATION

University of Michigan, Ann Arbor, MI — August 2025 - May 2030

PhD in Computer Science and Engineering

Focus: Agentic Systems, In-Network ML, and Domain-Specific LLMs

Purdue University, West Lafayette, IN — May 2025

Bachelor of Science, Computer Science GPA 3.9/4.0

Concentration in Machine Intelligence

Minor: Mathematics, Accounting, Management

Relevant Coursework: Theory of Computation, Data Mining and Machine Learning, Artificial Intelligence, Computer Architecture, Systems Programming

SUMMARY OF RESEARCH SKILLS

Project management • Data collection • Data analysis • Report writing • Research methodology & design • Data management • Oral presentations

RESEARCH INTERESTS

• Systems • Computer Networks • Machine Learning • Software Verification

RESEARCH EXPERIENCES

SRC Research Scholars Program — Feb 2025 – Present

Full-time Research Scholar, Ann Arbor, MI

- Conducting research on hardware-software co-design for wide-area distributed training of large language models (LLMs), focusing on optimizing communication efficiency and scalability across geographically dispersed data centers.

Purdue University, Department of Computer Science, NextGArchLab — Jan 2024 - Present

Undergraduate Research Assistant, West Lafayette, IN

Advisor: Dr. Muhammad Shahbaz

- Developing and optimizing ensemble machine learning models for network classification in PISA switches, with a focus on improving performance and accuracy.
- Implemented and benchmarked classification models (One-class SVMs, Random Forest, Isolation Forest) in Python to analyze packet payload features for malicious vs benign traffic, establishing baseline accuracy, computational cost, and scalability trade-offs for real-world security applications.
- Designing algorithms that enable efficient, data-driven decision-making in modern computer networks.

Purdue University, School of Electrical and Computer Engineering May 2024 - Present
Summer Undergraduate Research Fellowship, West Lafayette, IN
Advisor: Dr. Jenna DiVincenzo

- Evaluated open-source large language models (LLMs) for their ability to generate fully accurate specifications for static verification, applying techniques like prompt engineering, chain-of-thought (CoT), and few-shot prompting to assess model effectiveness.
- Categorized VeriFast benchmarks and developed tailored input datasets for training and testing LLMs.
- Designed qualitative analysis methodologies to quantify the quality of LLM outputs and conducted qualitative result interpretation for research and workshop publications.

Purdue University, Department of Chemistry May 2023 – August 2023
Undergraduate Research Assistant, West Lafayette, IN
Advisor: Dr. Tonglei Li

- Implemented a Python program for predicting 3-dimensional molecular structure and their properties based on user-input parameters in the project – "Deep Learning for Drug Discovery".
- Optimized the program using ML algorithms to decipher molecular mechanisms of crystal growth and improve prediction accuracy.

SELECTED PRESENTATIONS

TECHCON 2025, Semiconductor Research Corporation (SRC), Austin, TX, September 2025. Marilyn Rego, "SpliDT: Partitioned Decision Trees for Scalable Stateful ML Inference at Line Rate" (Talk and Poster)

Summer Undergraduate Research Symposium, Purdue University, West Lafayette, IN, July 2024. Marilyn Rego, "AI for Software Verification" (Oral Presentation)

PUBLICATIONS

Wen Fan*, **Marilyn Rego***, Xin Hu, Sanya Dod, Zhaorui Ni, Danning Xie, Jenna DiVincenzo (Wise), Lin Tan. *"Evaluating the Ability of GPT-4o to Generate Verifiable Specifications in VeriFast."* Accepted to **FORGE 2025** (*Co-First Author*).
https://arxiv.org/pdf/2411.02318

AWARDS AND HONORS

First-Year CSE PhD Rackham fellowship	2025
NSDI'25 Student Grant	2025
Dean's List and Semester Honors (All Semesters)	2021-2025
NSDI'26 Student Grant	2026

TEACHING & MENTORING EXPERIENCE

Undergraduate Research Mentor Fall 2025

NextGArch Lab, University of Michigan

- Mentoring two undergraduate students on research in ML and computer networks.
- Guiding them through experiment design, coding practices, and presenting research outcomes.

Undergraduate Teaching Assistant Spring 2023

CS 182 – Foundations of Computer Science

- Conducted weekly PSOs help sessions to enhance students' ability in mastering course concepts.
- Addressed students' inquiries regarding class materials and homework problems during weekly office hours, and online Q/A discussion platform.
- Evaluated and graded over 100 homework assignments biweekly and provided constructive feedback to enhance student learning and performance.

Peer Mentor Fall 2022 – Spring 2024

Purdue Global Science Partners (GSP)

- Provided support and direct resources to the needs of the mentees, which include academic, social, cultural or any other concerns.
- Ensured their success during their transition to college and their first year at Purdue University.
- Organized group activities and community events to promote intercultural awareness, and cultural diversity.

ORGANIZATIONS

VoiCSEs Officer, UMich Fall 2025

TECHNICAL SKILLS

Languages: C, Java, Python, HTML/CSS, JavaScript, R
Technologies/Environments : Windows, Linux, GitHub, Git, LaTeX